

Further Investigations on the Kinetics of Gaseous Oxidation Reactions

R. Fort and C. N. Hinshelwood

Proc. R. Soc. Lond. A 1930 **129**, 284-299

doi: 10.1098/rspa.1930.0156

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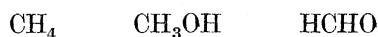
By R. FORT and C. N. HINSHELWOOD, F.R.S.

(Received June 28, 1930.)

Introduction.

Investigation of the kinetics of the oxidation of ethylene* and of benzene† showed that these reactions are peculiar in the following respects. First, the relation between the rate of reaction and concentration is such that the reactions possess no simple "order," though the nearest integral value for the order is about the third or fourth. The rate increases very rapidly with increasing hydrocarbon concentration, but is relatively little influenced by oxygen; under some conditions oxygen may have a retarding influence. Secondly, the reactions can be slowed down by increasing the surface exposed to the gases. This indicates that the oxidation occurs by a chain mechanism. Thirdly, the rate of change of pressure accompanying the oxidation only attains its full value after an induction period, during which evidently intermediate products are accumulating.

Accepting the fact that the oxidations are probably chain reactions, the relation between rate and concentration shows that the chains are much more easily propagated when the intermediate active molecules encounter more hydrocarbon than when they encounter oxygen. Following the view of Egerton,‡ and consistently with previous work on the combination of hydrogen and oxygen,§ the working hypothesis adopted is that some intermediate peroxidised substance is responsible for the propagation of the chains. This being so, the question arises whether the peculiarities found in the oxidation of hydrocarbons will also be found with substances already containing oxygen. To investigate, therefore, the influence of chemical configuration on the mechanism of oxidation reactions the following series of compounds|| has been studied



* Thompson and Hinshelwood, 'Proc. Roy. Soc.,' A, vol. 125, p. 277 (1929).

† Fort and Hinshelwood, 'Proc. Roy. Soc.,' vol. 127, p. 218 (1930).

‡ Egerton and Gates, 'J. Inst. Petroleum Technol.,' vol. 13, p. 281 (1927).

§ Thompson and Hinshelwood, 'Proc. Roy. Soc.,' A, vol. 122, p. 610 (1929).

|| Methane not in very great detail. We believe the problem of methane is also being studied elsewhere.

which represent the stages through which Bone and others have shown the oxidation of methane to occur.

It may be stated at the outset that the oxidations not merely of methane, but equally that of methyl alcohol and formaldehyde show the characteristics described above. Thus the first product of interaction of a hydrocarbon and oxygen is by no means the only one which can give rise to chains. Their propagation appears to be conditioned more by the energy available than by the exact chemical configuration of the molecules.

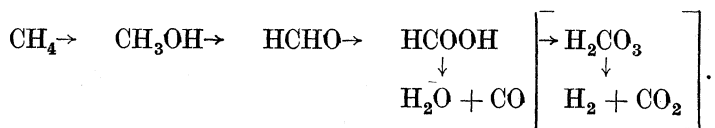
As will be seen, the kinetic relationships to which the chain mechanism gives rise are very complicated. But an approximate quantitative treatment of the typical example of methyl alcohol will be given in a later section.

Experimental Method.

The method of investigation was the same as that described in previous papers,* the reactions being followed by observation of the pressure changes which accompany them. Methane was obtained from a cylinder of natural gas freed from hydrogen by passage over copper oxide heated to 320°–340°, and then fractionated from liquid air after removal of carbon dioxide. It still contained some nitrogen, which was, however, considered to be harmless. Methyl alcohol vapour was obtained from a sample of the liquid which had been very highly purified for conductivity measurements. Formaldehyde vapour was obtained by heating dried paraformaldehyde in a small bulb attached to the reaction vessel.

The Course of the Reactions.

The principal products of reaction are carbon monoxide and steam, even when oxygen is in excess. According to Bone's scheme† this is because the final stage is the decomposition of formic acid



Only with methane is the proportion of carbon dioxide considerable, and

* 'J. Chem. Soc.,' vol. 125, p. 393 (1924); 'Proc. Roy. Soc.,' A, vol. 111, p. 245 (1926); vol. 118, p. 170 (1928); vol. 127, p. 218 (1930).

† "Flame and Combustion in Gases," Bone and Townend (1927).

here it is quite possible that some direct oxidation of carbon monoxide occurs at the rather high temperature required by the reaction.

	Methane. 150 mm. CH ₄ 300 mm. O ₂ .	Methyl alcohol. 100 mm. CH ₃ OH 200 mm. O ₂ .	Formaldehyde. 200 mm. HCHO 200 mm. O ₂ .
CO ₂	Per cent. 20·2	Per cent. 7·0	Per cent. 4·8
CO	29·2	35·2	54·0
O ₂	35·6	52·3	36·9
Residue	15·0	5·5	4·3

The range of temperature in which the rate of reaction is measurable is very different for the three substances, the temperature required being greatest for methane and least for formaldehyde.

	Temperature at which half life for standard pressure is 5 minutes.	Heat of activation.
CH ₄	470	Calories 61,500
CH ₃ OH	442	62,500
HCHO	321	20,600

As may be seen from the equations of the chemical changes, the increase of pressure accompanying the oxidation to carbon monoxide is great enough to serve as a measure of the progress of the reactions, *e.g.*, $\text{CH}_3\text{OH} + \text{O}_2 = \text{CO} + 2\text{H}_2\text{O}$ gives an increase equal to the pressure of methyl alcohol. Owing to the fact that some carbon dioxide is formed the observed pressure increase is somewhat smaller. As long as there was enough oxygen for combustion of all the methyl alcohol to carbon monoxide the pressure increase for a given pressure of the alcohol was constant and independent of the excess of oxygen. This fact facilitates the consideration of the results from the kinetic point of view.

Pressure of methyl alcohol.	Pressure of oxygen.	Increase.	Increase Pressure of CH ₃ OH
mm.	mm.		
50	250	42·5	0·85
100	250	85	0·85
150	250	130	0·87

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As shown in fig. 1, when the increase of pressure is plotted against time, the curve shows an induction period, the cause of which must be primarily connected with the occurrence of the oxidation in stages. Only when a small definite amount of the intermediate product has accumulated will its rate of destruction become equal to its rate of formation, and only when this stationary condition is established will the full rate of pressure increase be observed.

As an example an experiment with methyl alcohol may be considered. At 437° with 100 mm. methyl alcohol and 200 mm. oxygen the maximum rate of pressure increase was not attained for about 100 seconds. During this time the increase of pressure was 5 mm. From a curve it could be seen that had there been no induction period, that is, had the methyl alcohol been oxidised direct to carbon monoxide, the increase during this time would have been 14 mm., and this increase would represent the disappearance of 14 mm. of methyl alcohol. We may assume that these 14 mm. of methyl alcohol were in fact oxidised during the induction period, but only to intermediate products. If all had gone to formaldehyde the increase would have been 7 mm.; the observed increase is actually slightly less even than this, but we shall probably be right as regards order of magnitude if we conclude that at the end of the induction period most of the 14 mm. of methyl alcohol have been converted into formaldehyde. The question now is whether formaldehyde vapour at such a concentration would at this temperature be oxidised as fast as it is formed, so that a stationary condition can be established. By extrapolation of the direct measurements which were made it can be calculated that formaldehyde vapour would be oxidised at the required rate of 14 mm. per 100 seconds under the conditions of this experiment if its pressure were 17 mm. This is near enough to the concentration estimated to prevail at the end of the induction period for the establishment of the stationary condition to be quite possible.

There appears to be no difficulty about interpreting the induction period in the oxidation of methane in a similar way; but there is a difficulty in the case of formaldehyde. Here it would be natural to attribute the induction period to an initial accumulation of small amounts of formic acid. But formaldehyde is oxidised to carbon monoxide and water at a temperature where direct experiment under similar conditions showed formic acid to be very stable in comparison with what might have been expected. Thus the course of the oxidation must be somewhat more complex than the formulæ given at the beginning of this section indicate. While the scheme given there undoubtedly expresses the general behaviour correctly, other intermediate peroxidised substances probably play their part also.

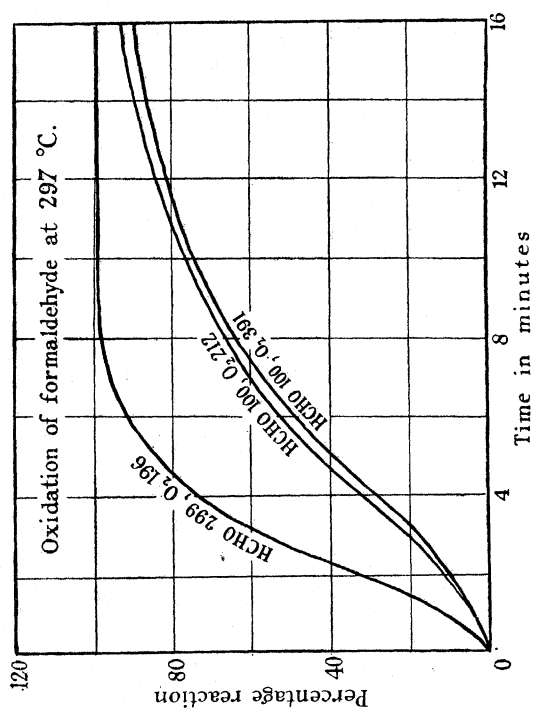
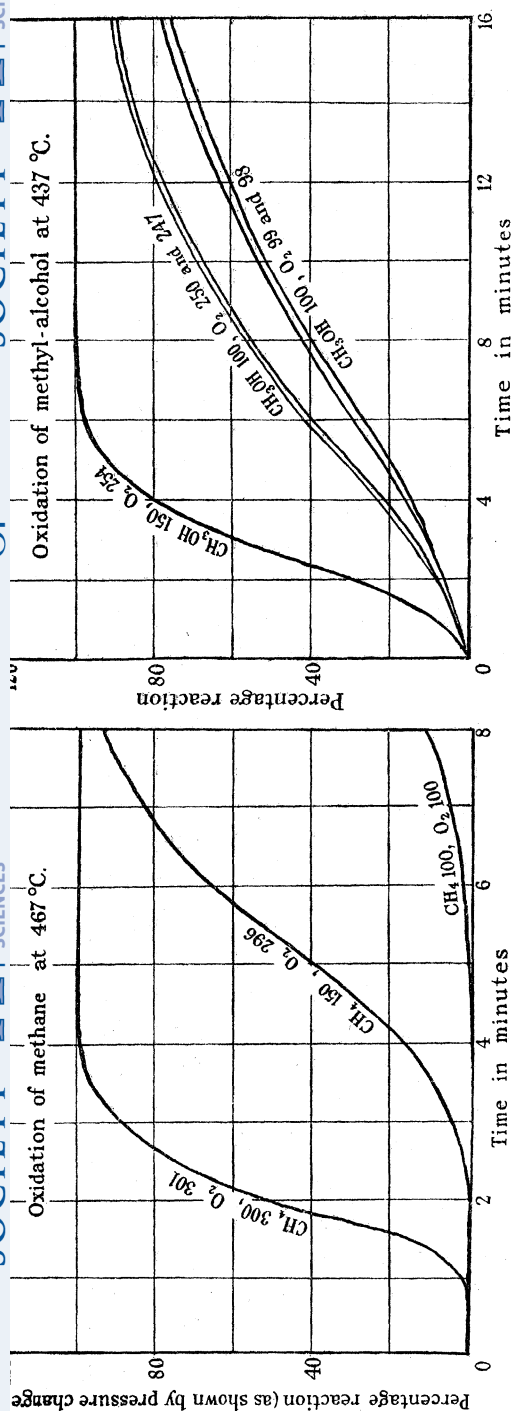


FIG. 1.

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Some typical experiments, from which the curves in fig. 1 are taken, are recorded below. t is the time, x is the percentage of the total pressure increase which has occurred.

Methane 467° C.

99 mm. CH ₄ . 202 mm. O ₂ . Total pressure increase 21 mm.		150 mm. CH ₄ . 296 mm. O ₂ . Total pressure increase 34 mm.		300 mm. CH ₄ . 301 mm. O ₂ . Total pressure increase 59 mm.	
t .	x .	t .	x .	t .	x .
' "		' "		' "	
6 58	5	3 6	6	1 1	2
7 52	10	3 53	15	1 16	7
9 26	19	4 22	23	1 34	19
11 16	33	4 52	35	1 45	31
12 35	52	5 14	44	1 55	42
16 10	67	5 33	53	2 6	56
19 40	85	6 0	65	2 21	71
		6 27	74	2 39	83
		6 58	82	2 58	88
		7 43	91	3 30	97
		8 31	97		

Methyl Alcohol 437° C.

100 mm. CH ₃ OH 99 mm. O ₂ . Total pressure increase 86 mm.		100 mm. CH ₃ OH 250 mm. O ₂ . Total pressure increase 84 mm.		150 mm. CH ₃ OH 254 mm. O ₂ . Total pressure increase 130 mm.	
t .	x .	t .	x .	t .	x .
' "		' "		' "	
1 36	3.5	1 32	4	0 36	3
2 24	7	2 15	8	1 0	7
3 23	13	3 45	20	1 26	14
5 8	24	4 22	26	1 53	26
6 53	36	6 18	44	2 33	41
8 46	48	7 46	56	2 55	57
11 5	60	9 35	68	3 13	64
14 6	72	11 57	80	3 34	72
16 19	81	13 53	86	4 4	80
21 24	91	17 31	95	5 5	91
23 33	94			6 51	98

Formaldehyde 297° C.

100 mm. HCHO. 212 mm. O ₂ . Total pressure increase 41 mm.		100 mm. HCHO. 391 mm. O ₂ . Total pressure increase 40 mm.		299 mm. HCHO. 196 mm. O ₂ . Total pressure increase 111 mm.	
<i>t.</i>	<i>x.</i>	<i>t.</i>	<i>x.</i>	<i>t.</i>	<i>x.</i>
' "		' "		' "	
1 2	4.5	1 25	5.5	0 27	3
2 6	12	2 5	10.5	0 44	7
2 57	20	3 14	21.5	1 14	16
4 9	34	4 29	35	1 39	25
5 33	49	5 38	46	2 2	35
7 8	63	6 32	54	2 47	53
8 12	71	7 41	65	3 13	62
11 49	85	10 47	78	4 29	80
17 13	95	18 42	95	6 16	94

Influence of Concentrations on the Rate of Reaction.

The analysis of the kinetics of these reactions is complicated by the existence of the induction period. We may, however, assume that when once the maximum rate of pressure increase is established it will measure the actual rate of oxidation of the initial substance, methane, methyl alcohol or formaldehyde as the case may be, since any intermediate products will be disappearing at the same rate as they are formed. Consequently in dealing with the kinetics of the various oxidations we have attached significance only to those parts of the curve which represent the reaction after the induction period. The method of recording the results depends upon the fact that, after the induction period is over, the ratio of the times required for any given fraction of the reaction at two pressures is approximately the same, whatever the actual fraction chosen may be. This is illustrated by the following table.

Initial pressure of		Relative times.				
CH ₃ OH.	O ₂ .	0-10 per cent.	10-20 per cent.	20-40 per cent.	40-60 per cent.	60-80 per cent.
150	250	1	1	1	1	1
50	250	4.8	9.5	10.3	10.4	12.4
		4.4	9.5	9.4	10.6	11.9

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Since the ratio of the times for each of the intervals after the first is approximately the same, the average values for the range 20 to 80 per cent. may be used for the purpose of expressing the influence of any desired factor on the rate of reaction. This method of representation is adopted in all the following tables.

Methane 467°.		Methyl alcohol 437°.		Formaldehyde 297°.	
(a) Oxygen pressure constant at 300 mm.		(a) Oxygen pressure constant at 250 mm.		(a) Oxygen pressure constant at 200 mm.	
Initial pressure of CH ₄ .	Relative values of <i>reciprocal</i> of time for reaction to go from 20-80 per cent.	Initial pressure of CH ₃ OH.	Relative reciprocal time.	Initial pressure of HCHO.	Relative reciprocal time.
100	1.00	50	1	50	1
200	2.98	100	3.09	100	1.48
300	4.94	150	9.84	150	2.02
				200	2.44
				300	3.15
				400	5.78
(b) Methane pressure constant at 200 mm.		(b) Methyl alcohol pressure constant at 100 mm.		(b) Formaldehyde pressure constant at 100 mm.	
Initial pressure of O ₂ .	Relative reciprocal time.	Initial pressure of O ₂ .	Relative reciprocal time.	Initial pressure of O ₂ .	Relative reciprocal time.
150	1.00	50	1	100	1.00
200	1.16	100	0.69	200	0.91
300	1.80	150	0.72	300	0.85
440	2.87	200	0.86	400	0.87
		250	0.93		
		300	0.85		
		400	1.11		

Every example given in the above table was duplicated, the average value being recorded. In all three cases the influence of the concentration of the combustible gas is very great, while except in the case of methane the influence of oxygen is small. The various relationships are illustrated in fig. 2.

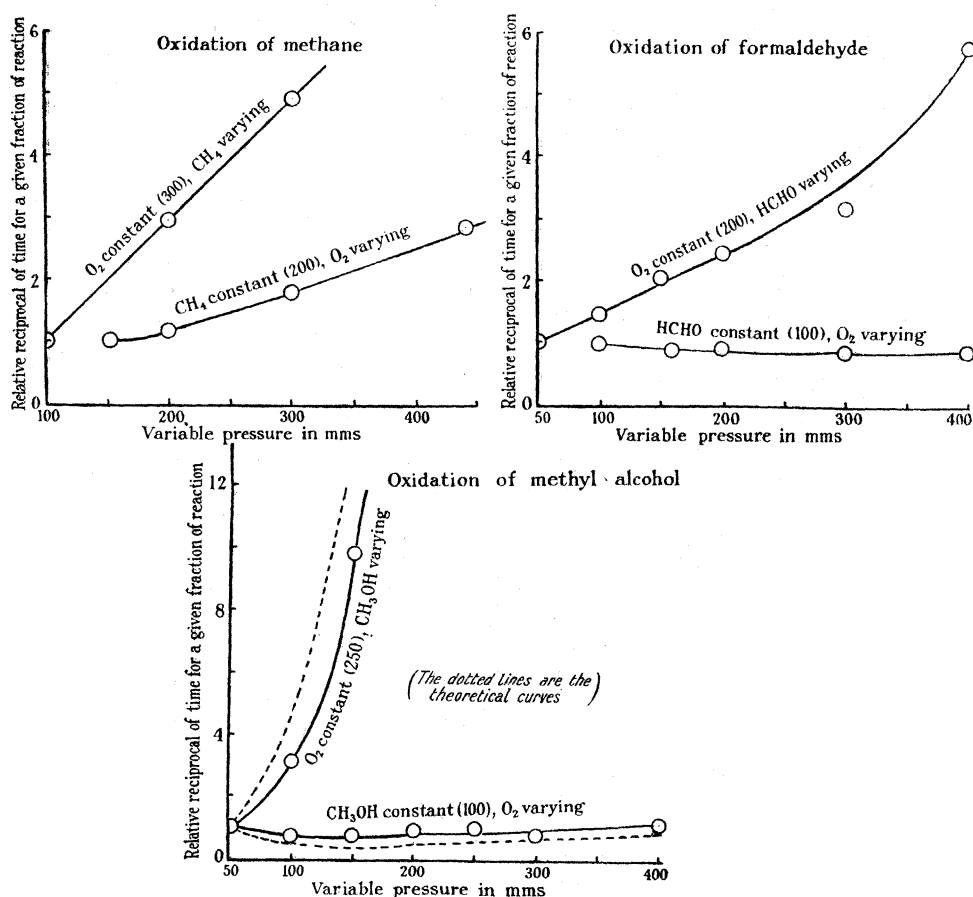


FIG. 2.

Influence of Surface.

If reaction chains start in the gas phase and end at the wall of the vessel, then the reaction will be much slower in a vessel of small than in one of large dimensions. But if the chains end mostly in the gas the size of the vessel will be unimportant. All three reactions here studied are slowed down considerably by a sufficient increase in the surface of the vessel.

With methane, for example, the following result is typical:—

150 mm. CH_4 : 298 mm. O_2 , 467°.
 Silica bulb. Time for 50 per cent. of total pressure increase
 5.5 minutes.

150 mm. CH_4 : 298 mm. O_2 , 467°.
 Similar silica bulb packed with silica tubes, 15 per cent. increase in 60 minutes. Analysis of gas withdrawn CO_2 , 0.4 per cent.; CO , 4.9 per cent.; O_2 , 61.5 per cent.; CH_4 , 33.2 per cent.

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In a vessel filled with small silica spheres the reaction would not proceed at all in 30 minutes under the above conditions, analysis of the gas withdrawn showing neither carbon monoxide nor dioxide.

Askey* has shown that the oxidation of formaldehyde, like that of benzaldehyde, occurs much more slowly in a bulb packed with silica powder than in an empty one. This was confirmed by the observation that in our bulb packed with silica spheres the reaction would hardly take place at 297°.

In the bulb packed with tubes the results with formaldehyde were very erratic. This fact, though disadvantageous from an experimental point of view, is very significant. It shows that the rate of reaction is governed not only by the extent of the surface and the dimensions of the vessel but by the qualitative nature of the wall. The variability of the latter factor is probably due to variations in the adsorbed gas layers. In this connection the following two facts are important. First, Pease† has shown that washing reaction vessels with a solution of potassium chloride markedly alters rates of oxidation. Secondly, it appeared in the course of this present series of experiments that the velocity of reaction in a new bulb often varied in the course of the first few experiments and then settled down to a constant value.

A more detailed investigation of the relation between the rate of reaction and the size of the vessel was made with methyl alcohol; the results are described in the following section.

Relation between Reaction Rate and Size of Vessel in the case of Methyl Alcohol. Comparison with the case of Benzene.

Two series of experiments were made. In the first series vessel A was of silica and of about 180 c.c. capacity, approximately cylindrical, length 10 cm., diameter 5.9 cm. Vessel B was also cylindrical, length 9.5 cm., diameter 6.7 cm., but being entirely packed with lengths of silica tubing of internal diameter 1 cm., offered only about one-sixth of the distance for the chains to travel before meeting the wall.

	Temperature 437° C.				100 mm. CH ₃ OH, 200 mm. O ₂ .			
Half life in vessel A (minutes)					3.5	3.5	3.5	3.9
Half life in vessel B (minutes)					17.8	14.2	16.6	17.6

These two bulbs had had very different previous histories. Therefore, a set

* 'J. Amer. Chem. Soc.,' vol. 52, p. 974 (1930).

† *Ibid.*, vol. 51, p. 1839 (1929).

of new bulbs of varying diameter but all of length 9 cm. and of the same silica were obtained. The following results were obtained with them.

100 mm. CH_3OH , 200 mm. O_2 .

Temperature.	Time required for reaction to go from 20 to 80 per cent. (minutes).			
	7.5 cm. diameter.	4.9 cm. diameter.	3.2 cm. diameter.	1.8 cm. diameter.
°				
427	6.6	25.5	59	Too long to measure. No experiment. (Anomalous 3.3.)
447	1.7	13.0	25.4	
463	0.80	4.8	14.6	

It can be shown that if reaction chains start in the gas and are broken whenever they reach the wall but never in the gas, then the rate of reaction should be approximately proportional to the square of the diameter of the vessel, if this is cylindrical and fairly long in comparison with its radius. The theory, however, leaves out of account possible qualitative differences between one vessel and another in regard to the efficiency with which the surface can break chains when they arrive there.

The above two series of results are not by any means in quantitative agreement with the simple theory, but in spite of this they show without any reasonable doubt that the chains must normally end on the walls, and thus may be of considerable length.

In a previous paper on the oxidation of benzene we concluded that the oxidation was only influenced by the dimensions of the vessel when these became very small (bulb packed with small spheres), and therefore that the chains were broken principally in the gas. The striking contrast between these results and the present results with methyl alcohol led us to repeat the benzene experiments with the four new bulbs.

Temperature 527° . 100 mm. benzene, 300 mm. oxygen.

Diameter of bulb (cm.)	7.5	4.9	3.2	1.8
Half life (minutes)	9.1	10.5	(20)	13.3

The existence of the very striking contrast in behaviour is thus confirmed.

The influence of nitrogen on the oxidation of methyl alcohol is to cause a

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small but definite acceleration of 30 to 40 per cent. for a pressure of nitrogen equal to that of the oxygen. The effect of a foreign gas is two-fold; it hinders the chains from reaching the walls, and it may itself deactivate. Thus the resultant effect is unpredictable.

Influence of Temperature.

In deriving the heats of activation the logarithm of the time required for the reaction to go from 20 to 80 per cent. was plotted against $1/T$ in the case of methane, while the logarithm of the half-life was plotted in the other two cases, since the influence of the induction period on the results is negligible.

Methane. CH ₄ 150 mm. O ₂ 300 mm.			Methyl alcohol. CH ₃ OH 100 mm. O ₂ 200 mm.			Formaldehyde. HCHO 100 mm. O ₂ 200 mm.		
Tempera- ture ° C.	<i>t</i> obs. (mins.).	<i>t</i> calc.	Tempera- ture ° C.	<i>t</i> obs. (mins.).	<i>t</i> calc.	Tempera- ture ° C.	<i>t</i> obs. (mins.).	<i>t</i> calc.
487	0.95	0.95	452	2.65	2.63	337	2.6	2.55
467	2.65	2.74	444	4.15	4.31	327	3.4	3.43
457	4.54	4.98	437	6.5	6.6	307	6.2	6.1
447	12.7	9.1	429	10.0	11.2	297	8.4	8.3
			422	18.9	17.4	287	11.3	11.5
			407	42.4	47.9			
E = 61,500 calories.			E = 62,500 calories.			E = 20,600 calories.		

$t_{\text{calc.}}$ is obtained from the Arrhenius equation using the value of E given in the table. The low value of E for formaldehyde is to be noted in connection with the low temperature at which the reaction takes place. Though no definite correlation can be expected in chain reactions, the influence of the exponential factor in which E occurs must always be pronounced.

Approximate Quantitative Treatment of the Oxidation of Methyl Alcohol.

The evidence of an earlier section shows that the chains start in the gas and are broken at the walls.

The other two facts on which the theory is based are the very marked influence of methyl alcohol concentration on the rate, and the small influence of oxygen. The first leads us to assume that chains are continued when an active intermediate substance, X , collides in the gas phase with more methyl

alcohol molecules ; the second indicates that oxygen has a deactivating effect which counteracts its normal influence as a reacting substance.

Since the chains are broken at the wall, we assume the deactivating influence of the oxygen to be exercised there ; that is, that the active molecules X are destroyed by reaction with adsorbed oxygen. This allows for the possibility of qualitative variations in the effectiveness of different surfaces.

Attention will be confined to the state of affairs prevailing after the induction period is over.

We have

$$\frac{-d[\text{CH}_3\text{OH}]}{dt} = k_1[\text{CH}_3\text{OH}][\text{O}_2] + k_2[\text{X}][\text{CH}_3\text{OH}], \quad (1) \quad (2)$$

$$\frac{d[\text{X}]}{dt_i} = k_1[\text{CH}_3\text{OH}][\text{O}_2] - k_2[\text{X}][\text{CH}_3\text{OH}] + k_2[\text{X}][\text{CH}_3\text{OH}] \quad (3) \quad (4) \quad (5)$$

$$-k_3 \frac{[\text{X}][\text{O}_2]^n}{[\text{CH}_3\text{OH}]^m} = 0. \quad (6)$$

(1) represents the initiation of the chain by a collision between methyl alcohol and oxygen forming X, and (2) its continuation by collision between X and methyl alcohol. (3) corresponds to (1) and (4) to (2) while (5) is introduced to express the fact that each X which disappears by reaction with methyl alcohol ultimately regenerates another X to continue the chain. (6) represents the breaking of the chain by oxidation of X on the wall. The rate of this depends on the amount of adsorbed oxygen, which will be a function of the oxygen concentration and of that of any competing gases, in this case of methyl alcohol itself. The two terms with the fractional powers n and m are intended to allow for this in a rough empirical way, but, as is obvious, the necessity for considering the adsorption equilibrium makes an exact treatment impossible. We will make a trial with m and n equal to unity for the sake of simplicity.

k_3 will be inversely proportional to the total pressure, since X diffuses less easily to the wall when the pressure is high.

Solving the equations and remembering that process (2) accounts for very much more of the reaction than process (1) we find

$$\frac{-d[\text{CH}_3\text{OH}]}{dt} = k[\text{CH}_3\text{OH}]^3 [\text{Total pressure}].$$

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If a and b are the initial amounts of methyl alcohol and oxygen, and x the amount of methyl alcohol oxidised at time t ,

$$\frac{dx}{dt} = k(a-x)^3(a+b+x),$$

the total pressure being $a+b+x$ since $\text{O}_2 + \text{CH}_3\text{OH} = \text{CO} + 2\text{H}_2\text{O}$.

$$kt = \frac{1}{(2a+b)} \cdot \frac{1}{2} \left\{ \frac{1}{(a-x)^2} - \frac{1}{a^2} \right\} + \frac{1}{(2a+b)^2} \left\{ \frac{1}{a-x} - \frac{1}{a} \right\} + \frac{1}{(2a+b)^3} \left\{ \log \frac{a}{a-x} + \log \frac{a+b+x}{a+b} \right\}.$$

The half life values are obtained by putting $x = a/2$ when oxygen is in excess and $x = b/2$ when methyl alcohol is in excess.

The results yielded by this equation are shown by the dotted curves in fig. 2. In broad outline the experimentally found relations are reproduced, but from the nature of the assumptions that have to be made about the adsorption relationships precise agreement cannot be expected.

No attempt was made to express mathematically the course of the curves in fig. 1. It may be remarked, however, that in a given experiment the rate does not fall off, as reaction proceeds, in as marked a manner as the influence of the initial pressure would lead one to anticipate. This may be due to the fact that the products of reaction displace oxygen from the surface and render it much more active. This effect would act in opposition to that of the falling methyl alcohol concentration. It is, of course, not allowed for in the above equation—which is therefore only applied to calculate the influence of the initial pressures.

Presumably consideration similar to the above apply to the other two oxidations, and it will be evident that from the number of factors which influence the rate of reaction the specific differences between one substance and another are easily understood, though unpredictable.

General Considerations.

Certain general conclusions emerge from these investigations which can be summarised as follows.

The much greater influence of hydrocarbon concentration than of oxygen concentration on the rate of reaction already found with ethylene and benzene appears to be general. In the theoretical treatment we have attributed this to a deactivating influence of oxygen.

All the reactions are chain reactions, but the length of chain and the method by which it is ended vary considerably from one reaction to another. With benzene, for example, the chains do not appear to reach the walls unless the free space in the vessel is very small, while with methyl alcohol nearly all the chains appear to end there.

The actual nature and condition of the wall have a considerable influence. With methyl alcohol, in order to account for the experimental results, we have been led to assume that the efficiency of the walls in breaking chains depends upon the layer of adsorbed oxygen.

The existence of a variety of methods by which chains are broken, and especially the probability that surface adsorption relations play a part accounts for the considerable divergences in the behaviour of different substances.

Incidentally, the very different extents to which the rate of oxidation of different substances is affected by the dimensions of the vessel may not be without practical significance in connection with the combustion of gaseous fuels.

Summary.

The peculiarities associated with gaseous oxidation reactions, such as that of ethylene or of benzene have been further investigated, with a view to discovering the part played by chemical configuration in determining the kinetics of the oxidation. The substances studied were methane, methyl alcohol and formaldehyde, the two latter being intermediate products in the oxidation of the former.

The following characteristics were common to all the reactions: an initial induction period, most pronounced with methane; the striking dependence of the rate of reaction on the concentration of the combustible gas, and the relatively small influence of oxygen concentration; the marked decrease in the rate of reaction in vessels of small dimensions.

These characteristics are best explained by assuming that the reactions occur by "chain" mechanism.

The operation of this mechanism does not seem to depend upon any particular chemical configuration of the reacting molecules, as long as the necessary energy is available.

The relation between rate and the dimensions of the vessel has been studied in detail for methyl alcohol, in the oxidation of which the chains appear to end only at the wall of the vessel, in contrast with the case of benzene where they end principally in the gas unless the vessel is very small.

The part played by the walls depends not only on the dimensions of the vessel but on the nature of the surface. It seems probable that the adsorbed layer of oxygen may play an important part in breaking the chains which reach the wall.

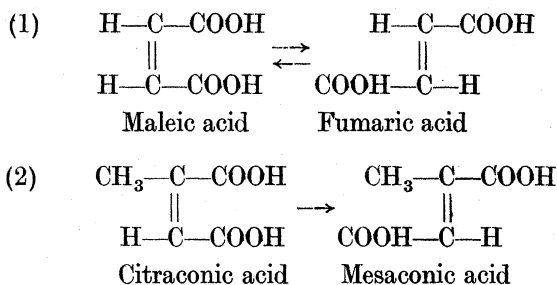
A theory based upon these assumptions gives results in general agreement with experiment.

Geometrical Inversion in Light.

By BHADRAMUKH K. VAIDYA, M.Sc. (Bombay).

(Communicated by E. C. C. Baly, F.R.S.—Received July 8, 1930.)

One of the most interesting types of photochemical change is exhibited by the class of organic compounds known as geometrical isomers. Under the influence of ultra-violet radiation these substances change from the *cis*-form to the *trans*-form and *vice versa*. Although many compounds of this class are known, no quantitative investigation has been made of the relation between the light energy absorbed and the amount of change produced except in the case of maleic and fumaric acids. This pair of geometrical isomers was investigated by Warburg,* who found a very low quantum efficiency for the change in both directions. Volmar,† however, stated that maleic-fumaric acids and citraconic-mesaconic acids underwent a decomposition of the same nature as oxalic and malonic acids, where carbon dioxide is one of the products of decomposition. The present investigation was undertaken to study the photochemical change in as many compounds of this type as possible, and the following pairs of isomers were examined :—



* 'Sitzungsber. Preuss. Akad. Wiss.,' vol. 50, p. 960 (1919).

† 'C. R.,' vol. 181, p. 467 (1925).